

Chapter 4 - Structure of Atom

Notes distilled from our learning conversation

The goal is **mastery and intuition**, not just definitions. The Maharashtra State Board Std XI textbook remains the primary sequence; the deeper experimental, historical and conceptual detours below come from our discussion.

Core idea of the chapter

This chapter is best understood as a chain of models being forced to change by experiments. Each model solved a real problem, but then created or exposed another one.

Thomson → Rutherford → Bohr → de Broglie + Heisenberg
→ Schrodinger → orbitals → electronic configuration

The deepest conceptual change is this:

Classical physics asks for the path of an electron. Quantum mechanics instead asks for the allowed quantum state of the electron and the probabilities that state predicts.

A recurring theme in these notes is to make hidden steps explicit: *what was actually measured, what was inferred, and which quantity was already known at each stage*. This matters especially in the Thomson, Chadwick, Bohr and quantum-mechanical parts of the chapter.

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1 Discovery of the electron: what Thomson actually established

Cathode rays were known before J. J. Thomson. In an evacuated discharge tube, a sufficiently large potential difference between electrodes produced a beam emerging from the cathode region.

The unresolved questions were more fundamental:

- Were cathode rays waves in a medium, or streams of particles?
- Did they carry electric charge?
- Were they produced by a particular gas or cathode material, or were they universal?
- If they were particles, what could be learned about their charge and inertia?

Thomson's key contribution was to combine electric and magnetic deflection experiments quantitatively. The cathode-ray carriers behaved like negatively charged particles, and essentially the same charge-to-mass ratio appeared with different gases and electrode materials. That made them universal constituents of matter: electrons.

1.1 Why the vacuum mattered

Earlier researchers could fail to observe electric deflection clearly even if the cathode rays were charged. Residual gas inside the tube could be ionized. The mobile positive and negative ions in that gas could rearrange and partially screen the applied electric field, making the field experienced by the cathode-ray beam much weaker than expected.

Better evacuation therefore did more than make the tube “cleaner”: it removed a physical effect that could hide the phenomenon being tested.

Experimental lesson

A failed observation need not mean an effect is absent. Sometimes the apparatus changes the very field or interaction one is trying to measure.

1.2 Electric deflection: what depends on what?

Suppose a particle of charge magnitude e and mass m moves horizontally with speed v through a uniform electric field E over a plate length l . The electric force is

$$F = eE,$$

so its vertical acceleration is

$$a_y = \frac{eE}{m}.$$

The time inside the plates is approximately

$$t = \frac{l}{v}.$$

Hence the vertical velocity gained is

$$v_y = a_y t = \frac{eEl}{mv}.$$

For a small angle,

$$\theta \approx \frac{v_y}{v} = \frac{eEl}{mv^2}.$$

This derivation assumes that the electric field is approximately uniform, that the horizontal speed does not change appreciably, and that the deflection angle is small.

It immediately shows why fast electrons are hard to deflect electrically:

$$\theta \propto \frac{1}{v^2}.$$

If the speed doubles, the electric deflection falls by a factor of four.

1.3 Where did Thomson get the speed from?

This is the key step that must be explicit, otherwise the argument looks circular.

In a transverse magnetic field B , the magnetic force magnitude is approximately

$$F_B = evB.$$

Over the same effective length l , the small magnetic deflection angle has the form

$$\phi \approx \frac{eBl}{mv}.$$

The electric deflection was

$$\theta \approx \frac{eEl}{mv^2}.$$

Take the ratio:

$$\frac{\phi}{\theta} = \frac{Bv}{E}.$$

Therefore

$$v = \frac{E}{B} \frac{\phi}{\theta}.$$

Notice what has happened: e and m cancel. Thomson can determine the beam speed from measurable field strengths and measurable deflections without knowing either the electron charge or the electron mass separately.

In his actual experiments he adjusted the magnetic and electric deflections to be equal in magnitude, so $\phi = \theta$. Then

$$v = \frac{E}{B}.$$

Once v is known, substitute it back into either deflection equation. For equal deflections, the electric equation gives

$$\theta = \frac{eEl}{mv^2} = \frac{eEl}{m(E/B)^2},$$

so

$$\frac{e}{m} = \frac{\theta E}{B^2 l}.$$

Everything on the right can be measured experimentally.

Logical order of the experiment:

$$E, B, \theta, \phi \implies v \implies \frac{e}{m}.$$

No prior value of e or m is needed.

1.4 What did the large e/m ratio really tell Thomson?

A very large e/m ratio by itself does not mathematically tell us whether the charge is extraordinarily large, the mass is extraordinarily small, or both. The experiment directly establishes only the ratio.

The decisive facts were that the ratio was far larger than for known ions and that it was reproducible across different cathodes and gases. Later measurements of the elementary charge, especially Millikan's oil-drop work, fixed e separately. Combining that with Thomson's e/m gave the electron mass.

Common trap

Do not write "Thomson measured the mass of the electron." He measured e/m . The separate mass became available only after the charge was independently known.

1.5 Useful speed estimate - label it as a later intuition, not Thomson's starting point

After e/m has been measured, a known accelerating potential can give a useful speed estimate. If a charged particle starts approximately from rest and gains kinetic energy by moving through a potential difference V ,

$$eV \approx \frac{1}{2}mv^2.$$

Rearranging without needing e and m separately,

$$v \approx \sqrt{2 \left(\frac{e}{m} \right) V}.$$

Using the modern value $e/m \approx 1.76 \times 10^{11} \text{ C kg}^{-1}$, an electron accelerated through 2000 V has

$$v \approx 2.65 \times 10^7 \text{ m s}^{-1} \approx 0.088c.$$

This is about 9 percent of the speed of light. It is a useful intuition for cathode-ray beams, but it must not be used as the logical step by which Thomson first obtained e/m ; that would be circular.

The quantity eV also explains the unit electron-volt: one electron passing through one volt gains one electron-volt of energy. A 2000 V accelerating potential corresponds to about 2000 eV = 2 keV per electron.

2 Thomson's atomic model

Ordinary matter is electrically neutral. Once electrons were established as negatively charged constituents of atoms, balancing positive charge had to exist somewhere in the atom.

Thomson proposed the plum-pudding model:

- positive charge distributed diffusely through the atom;
- electrons embedded in that positive background.

Why did this matter for scattering? A diffuse positive charge produces a comparatively gentle electric field at any one location. A fast heavy positive projectile should therefore accumulate many small deflections rather than encounter a tiny region capable of reversing its direction.

Rutherford's scattering experiment tested exactly this kind of prediction.

3 Rutherford's gold-foil experiment

A radioactive source emitted fast α -particles. The beam was narrowed and directed at a very thin gold foil. A zinc sulfide screen detected individual scattering events as tiny flashes.

The observations were qualitatively striking:

- most α -particles passed almost straight through;
- some were deflected through small angles;
- a very small fraction were deflected through large angles;
- a tiny number returned almost backward.

3.1 Why “most went straight through” implies mostly empty volume

A large deflection requires the projectile to pass close to a region producing a strong force. If such strong-scattering material filled most of the atom, large deflections would be common.

Instead, large deflections were rare. Therefore the region containing concentrated positive charge and mass must occupy only a tiny fraction of the atom's volume.

This is the logic behind the phrase

the atom is mostly empty space.

In modern language, the region outside the nucleus is not literally “nothing”: it contains the electron quantum state. The statement means that there is no massive, concentrated positive body filling the atomic volume.

3.2 Why the rare large deflections imply a nucleus

An α -particle is positively charged. A very large deflection requires a strong repulsive force over a small distance. Coulomb force has magnitude

$$F = k \frac{|q_1 q_2|}{r^2}.$$

Because of the $1/r^2$ dependence, a close approach to concentrated positive charge can produce an enormous force.

The large deflections therefore imply that the atom's positive charge is concentrated in a very small central region: the nucleus.

The nucleus must also contain nearly all of the atomic mass. A fast, heavy α -particle cannot be turned through a huge angle by exchanging momentum with a very light electron in the same way that a heavy ball is not easily bounced backward by striking a free, extremely light ball. The strong-scattering center must itself be massive.

3.3 The alpha particle does not have to “hit” the nucleus

Do not picture the large-angle event as two hard spheres making contact. The dominant interaction is electrostatic. The α -particle can approach the positive nucleus, slow in the inward direction as electrostatic potential energy rises, and reverse course without touching the nucleus.

3.4 How can a neutral atom repel a positive alpha particle so strongly?

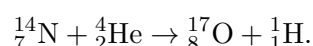
Electrical neutrality means the *total* positive and negative charges cancel when viewed from far away. It does not mean the electric field is zero everywhere inside the atom.

At large distances, the nuclear and electronic contributions mostly cancel. At very short distances near the nucleus, the concentrated positive nuclear charge dominates, producing strong repulsion.

This is an important general lesson: charge neutrality is a statement about the total charge, not about the absence of local electric fields.

4 The proton

Rutherford later bombarded nitrogen with α -particles and observed hydrogen nuclei among the products. A representative reaction is



Notice the nuclear bookkeeping:

$$14 + 4 = 17 + 1, \quad 7 + 2 = 8 + 1.$$

Mass number and nuclear charge are conserved in this reaction.

The hydrogen nucleus became identified as the proton, with charge $+e$ and mass close to 1 u.

Common trap

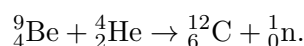
Do not picture this simply as an alpha particle mechanically knocking a pre-existing proton out of nitrogen. It is a nuclear transmutation: the nuclear constituents rearrange into new nuclei.

5 The neutron and Chadwick's detective work

Rutherford had proposed a neutral massive nuclear constituent. In 1932, James Chadwick established the neutron experimentally.

Beryllium bombarded with α -particles emitted a highly penetrating neutral radiation. Earlier workers had interpreted it as very energetic γ -radiation. A crucial observation was that this radiation could eject fast protons from paraffin wax.

A representative production reaction is



5.1 Why proton recoil reveals the mass of the unknown projectile

For an approximately elastic head-on collision between an incoming projectile of mass M and a stationary target of mass m , the target recoil speed is

$$V = \frac{2M}{M + m}u,$$

where u is the incoming projectile speed.

This equation gives useful intuition:

- If $M \ll m$, the heavy target barely moves.
- If $M \approx m$, the projectile can transfer a very large fraction of its kinetic energy to the target.

- If $M \gg m$, the light target can be kicked forward strongly, but the detailed recoil pattern differs.

Hydrogen nuclei are especially useful targets because the proton is light. The very energetic proton recoils were naturally explained if the incoming neutral particle had a mass comparable to the proton mass, so that an almost head-on collision could transfer a large fraction of the projectile's kinetic energy.

Chadwick also compared recoil behavior for more than one target nucleus. A single high-energy photon interpretation did not consistently account for the observed recoil energies, whereas a neutral massive particle did.

5.2 How can a neutral particle be detected?

A neutron is not deflected by electric or magnetic fields in the simple way a charged particle is, and it does not directly ionize matter as strongly as a charged projectile.

But a neutron can collide with a nucleus. The recoiling nucleus is charged and can ionize matter. Thus the detection chain can be

neutron $\rightarrow$ nuclear collision $\rightarrow$ charged recoil $\rightarrow$ ionization $\rightarrow$ detector signal.

We also discussed stopping range: a fast charged recoil gradually loses kinetic energy through ionization and excitation of matter. The distance it travels before stopping can therefore be calibrated to estimate its initial kinetic energy.

6 Atomic number, mass number and nuclide notation

The number of protons determines the chemical identity of an element:

Z = number of protons.

For a neutral atom,

number of electrons = Z .

If N is the number of neutrons, then

$$A = Z + N, \quad N = A - Z.$$

Nuclide notation is

$${}^A_ZX.$$

For example,



contains 18 protons, 18 electrons if neutral, and

$$40 - 18 = 22$$

neutrons.

7 Isotopes, isobars and isotones

Relationship	Same quantity	Different quantity
Isotopes	same Z	different A and neutron count
Isobars	same A	different Z
Isotones	same N	different Z and generally A

Examples:

$^{40}_{20}\text{Ca}$, $^{41}_{20}\text{Ca}$ isotopes

$^{40}_{18}\text{Ar}$, $^{40}_{20}\text{Ca}$ isobars

$^{39}_{19}\text{K}$, $^{40}_{20}\text{Ca}$ isotones, since both have $N = 20$.

Common trap

Isobars have the same *mass number*, not necessarily exactly the same measured atomic mass.

8 Why Rutherford's atom is classically unstable

Rutherford located the nucleus, but his model created a severe theoretical problem.

If an electron literally revolves around the nucleus, its velocity is continuously changing direction even if its speed is constant. A change in velocity means acceleration.

Classical electromagnetic theory says an accelerating charged particle radiates electromagnetic energy. Therefore a classical orbiting electron should

radiate $\rightarrow$ lose energy $\rightarrow$ orbit shrinks $\rightarrow$ spiral into nucleus.

But real atoms are stable.

Rutherford's model also did not explain the distribution and allowed energies of electrons, nor why atomic emission spectra contain discrete lines. These problems set the stage for Bohr.

9 Electromagnetic radiation: wavelength, frequency and photon energy

For electromagnetic radiation,

$$c = \nu\lambda,$$

where c is the speed of light, ν is ordinary frequency and λ is wavelength.

Thus

$$\nu = \frac{c}{\lambda}.$$

For a photon,

$$E = h\nu = \frac{hc}{\lambda}.$$

Therefore

$$\boxed{\lambda \downarrow \Rightarrow \nu \uparrow \Rightarrow E \uparrow}$$

for photons.

9.1 Do not confuse photon wavelength with electron matter wavelength

For a photon,

$$E = h\nu = \frac{hc}{\lambda}.$$

For a material particle such as an electron, de Broglie later gives

$$\lambda = \frac{h}{p}.$$

Both involve Planck's constant, but they describe different relationships.

10 Historical detour: why quantization first entered physics

Classical physics predicted too much black-body radiation at high frequencies, the ultraviolet catastrophe.

Planck found that the observed radiation law could be reproduced if energy exchange occurred in discrete amounts proportional to frequency:

$$E = h\nu.$$

A useful intuition is that higher-frequency modes require larger energy quanta. At a fixed temperature, very large quanta are difficult to populate thermally, preventing the classical runaway at high frequency.

The historical point is important: quantization first entered physics through a radiation problem before it became a theory of atomic structure.

11 Hydrogen line spectrum: the clue that energies are discrete

When electric discharge passes through gaseous hydrogen, the emitted light is not a continuous rainbow. It consists of discrete spectral lines.

If emitted light comes in photons with

$$E = h\nu,$$

then each observed frequency corresponds to a particular photon energy. A discrete set of photon energies strongly suggests a discrete set of atomic energy differences.

What the spectrum tells us - and what it does not yet tell us

The line spectrum alone does not reveal the exact structure of the atom. It tells us that the atom can release only certain energy differences. A model is still needed to explain what the allowed states are and why their energies take those values.

12 Balmer, Rydberg and wavenumber

Balmer discovered an empirical pattern for visible hydrogen lines. Rydberg generalized it:

$$\tilde{\nu} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right), \quad n_2 > n_1,$$

with

$$R_H \approx 109677 \text{ cm}^{-1}.$$

Here $\tilde{\nu}$ is **wavenumber**, not ordinary frequency:

$$\tilde{\nu} = \frac{1}{\lambda}.$$

Wavenumber commonly has units of cm^{-1} ; ordinary frequency has units of s^{-1} or Hz.

For emission, n_2 is the higher-energy starting level and n_1 is the lower-energy final level. Bohr later gave these integers a physical interpretation.

12.1 Hydrogen spectral series

Series	Final/lower level	Main region
Lyman	$n = 1$	ultraviolet
Balmer	$n = 2$	partly visible
Paschen	$n = 3$	infrared
Brackett	$n = 4$	infrared
Pfund	$n = 5$	infrared

13 Experimental detour: how spectral wavelengths were measured

A diffraction grating contains many closely and regularly spaced grooves. If the grating has N grooves per unit length, then the groove spacing is

$$d = \frac{1}{N}.$$

A narrow entrance slit selects a thin beam; a collimator makes the rays approximately parallel; the grating sends different wavelengths to different angles; a telescope or detector measures those angles.

For light observed at angle θ relative to the central direction, the path difference between rays from neighboring grooves is

$$\Delta = d \sin \theta.$$

Constructive interference occurs when

$$d \sin \theta = m\lambda,$$

where $m = 0, 1, 2, \dots$ is the diffraction order. Thus

$$\lambda = \frac{d \sin \theta}{m}.$$

The grating does not create the wavelengths; it spatially separates wavelengths already present in the source. The same wavelength can appear in multiple orders because different integer values of m can satisfy the condition at different angles.

13.1 Why crystals are natural diffraction structures for X-rays

Atomic plane spacings in crystals are of the same order as X-ray wavelengths. Visible wavelengths are much larger than typical atomic spacings.

For X-rays reflected from successive crystal planes, constructive interference gives Bragg's law:

$$2d \sin \theta = n\lambda.$$

Here d is the spacing between crystal planes, not the groove spacing of a visible-light grating. This detour connects spectroscopy to the later ability to probe atomic-scale structure directly.

14 Bohr's model of the hydrogen atom

Bohr retained a partly classical picture: an electron moving in a definite circular orbit around the nucleus. But he imposed nonclassical rules.

14.1 Stationary states

Only certain orbits are allowed. Each allowed orbit has a fixed radius and fixed energy. While an electron remains in such a stationary state, Bohr postulated that it does not radiate continuously.

This is not derived from classical electrodynamics. It is a new postulate introduced precisely because the classical prediction of continuous radiation gives unstable atoms.

14.2 Quantized angular momentum

Bohr imposed

$$mvr = n \frac{h}{2\pi} = n\hbar, \quad n = 1, 2, 3, \dots$$

Only orbits satisfying this condition are allowed.

14.3 Transitions

An electron changes between stationary states by absorbing or emitting a discrete amount of energy:

$$\Delta E = h\nu.$$

For a downward radiative transition, a photon is emitted. For an upward transition, energy must be supplied.

Common trap

The electron is not allowed to possess arbitrary energies between the allowed states. Also, a transition should not be pictured as a tiny ball spiraling continuously through every intermediate Bohr radius; the model assigns an initial state and a final state.

15 Where the Bohr radius and energy formulas come from

The formulas are easier to remember if their origin is visible.

For a hydrogen-like ion with nuclear charge $+Ze$, the Coulomb attraction provides the centripetal force for Bohr's circular orbit:

$$\frac{mv^2}{r} = k \frac{Ze^2}{r^2}.$$

So

$$mv^2 = k \frac{Ze^2}{r}.$$

Bohr also requires

$$mvr = n\hbar.$$

Using the angular-momentum condition to eliminate v gives

$$r_n = \frac{n^2 \hbar^2}{mkZe^2}.$$

For hydrogen, $Z = 1$, and the constant part is the Bohr radius

$$a_0 = 52.9 \text{ pm},$$

so

$$r_n = n^2 a_0.$$

For a hydrogen-like species,

$$r_n = 52.9 \frac{n^2}{Z} \text{ pm}.$$

15.1 Why the total energy is negative

The kinetic energy is

$$K = \frac{1}{2}mv^2.$$

From the force relation,

$$mv^2 = k \frac{Ze^2}{r},$$

so

$$K = \frac{1}{2}k \frac{Ze^2}{r}.$$

The electrostatic potential energy of opposite charges is

$$U = -k \frac{Ze^2}{r}.$$

Therefore the total energy is

$$E = K + U = \frac{1}{2}k \frac{Ze^2}{r} - k \frac{Ze^2}{r} = -\frac{1}{2}k \frac{Ze^2}{r}.$$

Substituting r_n gives

$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J}$$

or for hydrogen

$$E_n = -\frac{13.6}{n^2} \text{ eV}.$$

This also explains why the energy is negative. The kinetic energy is positive, but the attractive potential energy is negative and twice as large in magnitude, leaving a negative total.

The zero of energy is chosen for a free electron infinitely far from the nucleus and at rest:

$$E_\infty = 0.$$

A bound state lies below that reference. For ground-state hydrogen,

$$E_1 = -13.6 \text{ eV},$$

so 13.6 eV must be supplied to ionize one atom. Per mole this is about

$$1312 \text{ kJ mol}^{-1}.$$

As $n \rightarrow \infty$, $E_n \rightarrow 0^-$: the ionization limit.

16 How Bohr explains the hydrogen spectrum

For emission from n_i to n_f ,

$$\Delta E = E_i - E_f > 0.$$

The emitted photon satisfies

$$h\nu = \Delta E.$$

Substituting the Bohr energies gives

$$\tilde{\nu} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

Thus the empirical Rydberg integers become labels of allowed energy states.

16.1 Worked example: $n = 5 \rightarrow n = 2$

$$\tilde{\nu} = 109677 \left(\frac{1}{4} - \frac{1}{25} \right) \text{ cm}^{-1} \approx 23032 \text{ cm}^{-1}.$$

Then

$$\lambda = \frac{1}{\tilde{\nu}} \approx 4.34 \times 10^{-5} \text{ cm} = 434 \text{ nm}.$$

It is a Balmer line because the final state is $n = 2$.

Common trap

The emitted object is a photon. Use $E = h\nu = hc/\lambda$. Do not equate the photon energy to an electron's ordinary kinetic energy and then use the electron de Broglie relation.

17 What exactly is emitted in an atomic transition?

It is misleading to imagine a photon stored inside the excited electron. The atom begins in a higher-energy state. During a radiative transition, the atomic state changes and a photon is created with

$$E_{\gamma} = \Delta E.$$

An excited atom may also relax through intermediate states, producing a cascade of several photons. Some relaxation can be non-radiative, with energy transferred through collisions or other processes.

18 Limitations of the Bohr model

Bohr's theory is remarkably successful for hydrogen and other one-electron species, but it is not a complete theory of atoms. It cannot satisfactorily explain

- spectra of multi-electron atoms;
- fine structure of hydrogen spectra;
- Zeeman splitting in magnetic fields;
- Stark splitting in electric fields;
- chemical bonding;
- the full wave-particle behavior of electrons.

Its deepest conceptual weakness is the definite classical orbit. Even for hydrogen, modern quantum mechanics does not say that the electron travels on a sharply defined circular path.

Bohr got much of the hydrogen energy spectrum right, but the literal orbit picture is not the modern physical description.

19 de Broglie: matter has wave character

Light had displayed both wave-like and particle-like behavior. de Broglie proposed that matter should also have wave character:

$$\lambda = \frac{h}{p}.$$

For a nonrelativistic particle, $p = mv$, so

$$\lambda = \frac{h}{mv}.$$

A numerical scale helps. For an electron moving at 10^6 m s^{-1} ,

$$\lambda \approx \frac{6.626 \times 10^{-34}}{(9.11 \times 10^{-31})(10^6)} \approx 7.3 \times 10^{-10} \text{ m} = 0.73 \text{ nm}.$$

That is comparable to atomic dimensions, so wave effects are not a tiny correction for electrons.

For a macroscopic object, the same formula gives an unimaginably small wavelength because the momentum is enormous. Electron diffraction later confirmed de Broglie's prediction.

20 A useful but limited bridge: standing waves and Bohr's rule

If we temporarily keep Bohr's circular orbit and require the matter wave to fit around the circumference without phase mismatch,

$$2\pi r = n\lambda.$$

Using

$$\lambda = \frac{h}{mv}$$

gives

$$2\pi r = n \frac{h}{mv}$$

and hence

$$mvr = n \frac{h}{2\pi}.$$

This reproduces Bohr's angular-momentum rule.

Common trap

This is an intuitive bridge, not the modern derivation of atomic orbitals. Quantum mechanics does not begin with a literal electron wave wrapped around a pre-existing circular path.

21 Heisenberg uncertainty principle

The uncertainty principle states

$$\Delta x \Delta p_x \geq \frac{\hbar}{2} = \frac{h}{4\pi}.$$

Here Δx and Δp_x are spreads in the quantum state's position and momentum distributions. They are not simply the calibration errors of measuring instruments.

21.1 Why localizing an electron creates momentum spread

Suppose an electron has exact momentum p_0 . de Broglie gives one exact wavelength:

$$\lambda_0 = \frac{h}{p_0}.$$

A pure single-wavelength state is spatially delocalized; it does not pick out a small region where the electron is concentrated.

To make a localized wave packet, many wavelengths must be superposed:

$$\lambda_1, \lambda_2, \lambda_3, \dots$$

But since

$$p = \frac{h}{\lambda},$$

many wavelengths imply a range of momenta:

$$p_1, p_2, p_3, \dots$$

Therefore tighter localization requires a broader momentum distribution:

$$\Delta x \downarrow \Rightarrow \Delta p \uparrow.$$

Exact momentum means one exact wavelength, not many wavelengths. The mixture of wavelengths appears when we try to localize the electron.

21.2 Why this destroys a classical orbit

A classical orbit assumes that, at every instant, the electron has a definite position and definite momentum. Quantum mechanics does not permit such a state for an atomic electron.

So the question

Where exactly is the electron, and what exact path is it following?

is not the right fundamental question.

22 Schrodinger's quantum-mechanical model

Schrodinger introduced

$$\hat{H}\psi = E\psi.$$

Here $\hat{H}$ is the Hamiltonian operator representing total energy, ψ is the wavefunction and E is the energy associated with that state.

22.1 What does an equation of this form mean?

This is an eigenvalue equation. The operator $\hat{H}$ acts on a trial function ψ . Most arbitrary functions do not represent acceptable stationary states. We seek special functions for which the result is the same function multiplied by a number E .

A useful analogy is a vibrating string fixed at both ends. Not every wavelength fits the boundary conditions. Only standing waves that match the endpoints are allowed. Their frequencies are discrete.

Similarly, the Schrodinger equation plus physical boundary conditions - the wavefunction must be single-valued, finite and normalizable - selects only certain acceptable solutions. Quantized energies emerge from the allowed solutions rather than being inserted as an extra Bohr rule.

Why this is a conceptual advance over Bohr

Bohr says: "Only certain energies are allowed" as a postulate. Schrodinger says: "Solve the wave equation with the physical boundary conditions; only certain energies survive."

22.2 Hydrogen: what Schrodinger keeps and what he discards

For hydrogen, the Schrodinger equation gives the characteristic spectrum

$$E_n \propto -\frac{1}{n^2},$$

matching Bohr's allowed energies in the nonrelativistic approximation. But it does not produce circular electron trajectories. The successful energy spectrum survives; the orbit picture is discarded.

22.3 What about multi-electron atoms?

The framework applies to multi-electron atoms and molecules, but exact analytical solutions generally become impossible because electron-electron repulsion couples the particles together.

Strictly speaking, the exact many-electron wavefunction depends on the coordinates of all electrons simultaneously. The familiar one-electron orbitals used in chemistry are extremely useful building blocks and approximations for describing that many-electron problem.

Practical quantum chemistry therefore uses approximations and numerical methods rather than an exact closed-form solution for every atom and molecule.

23 Wavefunction and probability density

The wavefunction ψ should not be imagined as an ordinary material wave like a vibrating rope.

The measurable spatial probability information comes from

$$|\psi(\mathbf{r})|^2.$$

This is a **probability density**. The probability of finding the electron inside a small volume element dV is

$$dP = |\psi(\mathbf{r})|^2 dV.$$

The wavefunction is normalized so that the total probability over all space is 1:

$$\int |\psi|^2 dV = 1.$$

This distinction prevents a common wording error: $|\psi|^2$ is not “the probability at an exact mathematical point”; it is a density whose integral over a region gives a probability.

24 Atomic orbitals: state, not path

An atomic orbital is an allowed one-electron wavefunction/state in an atom. It is not a trajectory.

Bohr orbit	definite circular path of an electron around the nucleus
Quantum orbital	allowed wavefunction/state describing energy and spatial probability distribution

The familiar orbital pictures are usually boundary surfaces chosen to enclose a large fraction of the probability. They are not physical containers.

25 Where the quantum numbers come from

Solving hydrogen in three-dimensional space is easiest in spherical coordinates (r, θ, ϕ) because the Coulomb potential depends only on distance from the nucleus.

The wavefunction can be separated schematically as

$$\psi(r, \theta, \phi) = R(r) \Theta(\theta) \Phi(\phi).$$

The requirement that these pieces be physically acceptable produces discrete labels.

25.1 Principal quantum number n

The radial equation and its boundary/normalization conditions produce

$$n = 1, 2, 3, \dots$$

The principal quantum number labels shells and, for hydrogen, determines the energy. In multi-electron atoms, energy is not determined by n alone.

Traditional shell labels are

$$1 \rightarrow K, \quad 2 \rightarrow L, \quad 3 \rightarrow M, \quad 4 \rightarrow N.$$

A shell with principal quantum number n contains

$$n^2$$

orbitals and can accommodate at most

$$2n^2$$

electrons.

25.2 Azimuthal quantum number l

The angular equation produces

$$l = 0, 1, 2, \dots, n - 1.$$

These correspond to subshells

$$0 \rightarrow s, \quad 1 \rightarrow p, \quad 2 \rightarrow d, \quad 3 \rightarrow f.$$

The value of l is tied to the angular shape and angular momentum properties of the orbital.

25.3 Magnetic quantum number m_l

The azimuthal angular dependence must return to the same physical value after a full rotation:

$$\Phi(\phi + 2\pi) = \Phi(\phi).$$

This periodicity forces an integer label. For a given l ,

$$m_l = -l, -l + 1, \dots, 0, \dots, l.$$

Therefore a subshell contains

$$2l + 1$$

orbitals:

Subshell	l	Number of orbitals
s	0	1
p	1	3
d	2	5
f	3	7

25.4 Spin quantum number m_s

Electron spin introduces

$$m_s = +\frac{1}{2} \quad \text{or} \quad -\frac{1}{2}.$$

This fourth quantum number does not emerge from the ordinary nonrelativistic spatial Schrodinger equation in the same way as n, l, m_l . Spin is an intrinsic quantum property and should not be pictured literally as a tiny charged sphere rotating around its axis.

Thus an electron in the orbital model is labeled by

$$\boxed{(n, l, m_l, m_s)}.$$

26 Example: derive the entire $n = 3$ shell

For $n = 3$,

$$l = 0, 1, 2.$$

Therefore the shell contains

$$3s, \quad 3p, \quad 3d.$$

The orbital counts are

$$1, \quad 3, \quad 5,$$

so the total is

$$1 + 3 + 5 = 9 = 3^2.$$

With two electrons per orbital, the maximum number of electrons is

$$2 \times 9 = 18 = 2n^2.$$

Common trap

$n = 3$ is a shell. $3s$, $3p$ and $3d$ are subshells.

27 A subtle point about p_x, p_y, p_z

For a p subshell,

$$m_l = -1, 0, +1.$$

It is tempting to identify these directly with p_x, p_y, p_z , but that is not strictly correct. The $m_l = 0$ state naturally has the angular form associated with p_z . The familiar real p_x and p_y orbitals are linear combinations of the $m_l = \pm 1$ complex eigenfunctions.

For this course, the practical idea is simply that a p subshell contains three differently oriented orbitals of the same energy when no external field breaks the degeneracy.

28 Shapes of orbitals and nodes

28.1 s orbitals

s orbitals are spherically symmetric. Their characteristic spatial extent generally increases with n .

28.2 p orbitals

p orbitals have two lobes separated by a nodal plane through the nucleus. The three p orbitals differ mainly by orientation.

28.3 d orbitals

A d subshell contains five orbitals with more complicated angular shapes. Four are commonly drawn with four lobes; d_{z^2} has a distinctive two-lobe-plus-torus form.

28.4 What is a node?

A node is a region where

$$\psi = 0,$$

so

$$|\psi|^2 = 0.$$

The probability density for detecting the electron there is zero.

For hydrogen-like orbitals:

$$\text{total nodes} = n - 1,$$

$$\text{angular nodes} = l,$$

$$\text{radial nodes} = n - l - 1.$$

Thus $2s$ has one radial spherical node, while $2p$ has one angular node, a plane.

Common trap

A node is not a physical wall that a little electron ball must cross. Quantum mechanics is not assigning a classical trajectory through the orbital.

29 Orbital drawings do not show hard walls

The probability density generally extends beyond the boundary drawn in a textbook diagram. The boundary is conventionally chosen to enclose a large fraction, often around 90 percent, of the total probability.

Outside the boundary, the probability becomes smaller but does not abruptly vanish.

Orbital picture = probability visualization, not a container.

30 Orbital energies in hydrogen

Hydrogen has only one electron, so there is no electron-electron repulsion or shielding by other electrons.

For the ideal nonrelativistic Coulomb problem, the energy depends only on n :

$$1s < 2s = 2p < 3s = 3p = 3d < \dots$$

Orbitals with different wavefunctions but the same energy are degenerate.

Thus $2s$ and $2p$ are degenerate in ideal hydrogen even though their shapes are different. Equal total energy does not imply identical spatial distributions.

31 Multi-electron atoms: shielding and penetration

Once several electrons are present, each electron experiences both attraction to the nucleus and repulsion from the other electrons.

Inner electrons partially shield outer electrons from the full nuclear charge. We often summarize the net attraction using an effective nuclear charge Z_{eff} .

31.1 What “shielding” really means

Suppose an outer electron is far from the nucleus. Much of the time, inner electrons lie between it and the nucleus. Their negative charge partially offsets the positive nuclear attraction felt by the outer electron.

Shielding is therefore not a rigid physical screen. It is a consequence of the average electron distribution and electron-electron repulsion.

31.2 What “penetration” really means

Penetration does not mean that the entire s orbital is simply “closer” to the nucleus than the entire p orbital.

It means that an orbital has appreciable probability density inside regions that are also occupied by inner shells, where the electron experiences less shielding and more of the nuclear charge directly.

Near the origin, the radial behavior is qualitatively

$$R(r) \propto r^l.$$

Therefore s orbitals can have substantial density very near the nucleus, while p , d and f orbitals are increasingly suppressed there. The qualitative penetration order is

$$s > p > d > f.$$

Greater penetration usually means larger Z_{eff} and lower energy within the same general shell region. A subtle but important example is $2s$ versus $2p$. The $2s$ orbital has an outer region that extends far from the nucleus, yet it also has nonzero probability close to the nucleus. That inner penetration can stabilize it relative to $2p$ in a multi-electron atom.

31.3 Why hydrogen is special

Hydrogen has no other electron to provide shielding. Its pure Coulomb potential has a special degeneracy, so $2s$ and $2p$ have the same total energy in the ideal nonrelativistic treatment even though their kinetic and potential-energy distributions differ.

32 The $n + l$ rule and orbital filling order

A practical rule for many-electron atoms is:

1. Lower $n + l$ generally means lower orbital energy.
2. If two orbitals have the same $n + l$, the one with lower n is usually lower.

For $3p$ and $4s$,

$$3p : n + l = 3 + 1 = 4, \quad 4s : n + l = 4 + 0 = 4.$$

Tie, so lower n gives

$$3p < 4s.$$

For $3d, 4p, 5s$, all have $n + l = 5$, so

$$3d < 4p < 5s.$$

A common practical filling sequence is

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < \dots$$

Why this does not contradict later $4s/3d$ behavior

The $n + l$ rule is a useful ordering guide for constructing many ground-state configurations. Orbital energies are not immutable labels independent of occupancy. When electrons are added or removed, especially in transition metals, the relative $4s$ and $3d$ energies can change.

33 Can hydrogen occupy states such as $5d$ or $6f$?

Yes.

For $5d$,

$$n = 5, \quad l = 2.$$

For $6f$,

$$n = 6, \quad l = 3.$$

Both are valid excited hydrogen states.

As n increases, the energy approaches zero from below, the state becomes more weakly bound, and its characteristic size grows roughly as n^2 . Very high- n states are Rydberg states.

Not every state can be reached by every one-photon transition. Ordinary electric-dipole transitions obey the selection rule

$$\Delta l = \pm 1.$$

Thus a direct one-photon $1s \rightarrow 5d$ transition is not an ordinary electric-dipole-allowed transition. Collisions, cascades and multiphoton processes can nevertheless populate higher states.

34 Aufbau principle

“Aufbau” means building up. To construct a ground-state electronic configuration, electrons are assigned to orbitals so as to build the low-energy arrangement, using the approximate orbital-energy order.

This is a bookkeeping and energy-minimization principle. It should not be interpreted too literally as saying that an atom in nature physically captures electrons one by one in exactly that historical sequence.

A common practical order is

$$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow \dots$$

Energy order alone is not sufficient; Pauli and Hund determine how electrons occupy available orbitals.

35 Pauli exclusion principle

Pauli’s principle states:

No two electrons in the same atom can have the same set of all four quantum numbers.

For two electrons in the same orbital, n, l, m_l are already identical. Therefore their spins must differ:

$$m_s = +\frac{1}{2}, \quad m_s = -\frac{1}{2}.$$

So one orbital can contain at most two electrons, with opposite spins.

36 Hund’s rule of maximum multiplicity

When several degenerate orbitals are available in a subshell, electrons occupy them singly with parallel spins before pairing begins.

For a p subshell:

Configuration	Orbital occupancy
p^1	$\uparrow \quad \square \quad \square$
p^2	$\uparrow \uparrow \quad \square \quad \square$
p^3	$\uparrow \uparrow \uparrow \quad \square \quad \square$
p^4	$\uparrow \downarrow \quad \uparrow \uparrow \quad \square$
p^5	$\uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow$
p^6	$\uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow$

Thus nitrogen $2p^3$ has 3 unpaired electrons, oxygen $2p^4$ has 2, and fluorine $2p^5$ has 1.

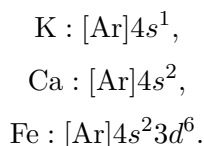
A useful physical intuition is that forcing two electrons into the same orbital pairs them in the same spatial region. When equal-energy orbitals are still empty, distributing electrons among them can lower the total energy; the full quantum explanation involves exchange as well as electrostatic repulsion.

Pauli answers: how can two electrons coexist in one orbital? Hund answers: which equal-energy orbital gets the next electron before pairing begins?

37 Electronic configuration and condensed notation

Electronic configuration is the distribution of electrons among the available orbitals.

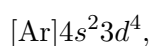
Examples:



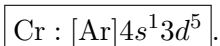
Condensed noble-gas notation replaces the fully filled inner core with the symbol of the preceding noble gas.

38 Chromium and copper: important exceptions

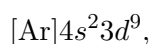
A naive Aufbau construction gives chromium



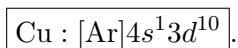
but the observed ground state is



For copper, the naive pattern suggests



while the observed ground state is



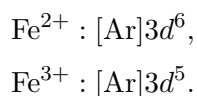
At textbook level, these are described using extra stability of half-filled and completely filled d subshells. The deeper caveat is that $4s$ and $3d$ are close in energy and the total energy depends on electron-electron repulsion, exchange and correlation. “Half-filled/full-filled stability” is a useful mnemonic, not a magical universal rule.

39 Transition-metal ions: why $4s$ electrons are removed first

Neutral iron is



But



The $4s$ electrons are removed before the $3d$ electrons.

This seems contradictory only if we assume a permanent orbital-energy ranking. In reality, orbital energies depend on the electron population. Once $3d$ is occupied, the $4s$ electrons are generally more spatially extended and more weakly bound in the neutral transition atom. Ionization therefore removes them first.

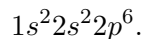
Filling order and ionization order are not necessarily exact reverses of one another.

40 Isoelectronic species and size

Species with the same number of electrons are isoelectronic. Consider



Each has 10 electrons:



Their nuclear charges increase from 8 to 12. Because the electron count and broad shielding pattern are similar, increasing nuclear charge pulls the same ten-electron cloud inward more strongly.

Therefore

$$\boxed{\text{O}^{2-} > \text{F}^{-} > \text{Ne} > \text{Na}^{+} > \text{Mg}^{2+}}$$

where $>$ means larger radius. Equivalently, smallest to largest:

$$\boxed{\text{Mg}^{2+} < \text{Na}^{+} < \text{Ne} < \text{F}^{-} < \text{O}^{2-}}.$$

Different neutron counts are not the important cause of this trend; the dominant variable is nuclear charge.

40.1 Does larger nuclear charge change the orbital shape?

The basic angular character remains: an s orbital remains spherical and a p orbital remains p -like. The main change is radial contraction:

$$Z \uparrow \Rightarrow \text{stronger attraction} \Rightarrow \text{more compact electron distribution.}$$

41 Selected mastery problems and what they taught us

The mistakes during practice were useful because they exposed exactly where a formula could be used without its physical meaning.

41.1 Rutherford reasoning

The rarity of large-angle events tells us the strongly scattering region occupies a tiny fraction of atomic volume. The magnitude of the deflection tells us positive charge and mass are highly concentrated there.

Small and large α deflections are primarily due to Coulomb interaction with the positive nucleus, not attraction to much lighter electrons.

41.2 Hydrogen $5 \rightarrow 2$ transition

Three different quantities must be kept separate:

- photon frequency ν ;
- photon wavelength λ ;
- spectroscopic wavenumber $\tilde{\nu} = 1/\lambda$.

The deeper lesson was not to mix photon relations with electron matter-wave relations.

41.3 Quantum-number detective

Allowed sets satisfy

$$\begin{aligned}n &= 1, 2, 3, \dots, \\l &= 0, 1, \dots, n-1, \\m_l &= -l, \dots, +l, \\m_s &= \pm \frac{1}{2}.\end{aligned}$$

Thus $(3, 2, -2, +\frac{1}{2})$ is allowed, while $(3, 3, 0, -\frac{1}{2})$ is not because $l < n$ must hold.

41.4 Orbital-energy ordering

For the set we practiced,

$$3p < 4s < 3d < 4p < 5s.$$

The $n + l$ rule plus the lower- n tie-breaker gives this cleanly.

41.5 Hund's rule and unpaired electrons

The correct unpaired counts are

$$\text{N} : 3, \quad \text{O} : 2, \quad \text{F} : 1.$$

The error to avoid is pairing before every equal-energy orbital has one electron.

41.6 Chromium integrated problem

For neutral chromium, $Z = 24$.

Naive Aufbau:

$$[\text{Ar}]4s^23d^4.$$

Actual ground state:

$$[\text{Ar}]4s^13d^5.$$

It has six unpaired electrons in total: five in $3d$ and one in $4s$.

One possible set of quantum numbers for a $3d$ unpaired electron is

$$(3, 2, 0, +\frac{1}{2}).$$

42 Common conceptual traps from Chapter 4

1. **Thomson's measurement becomes circular if speed is left unexplained.** Speed comes from combining electric and magnetic deflections so that the unknown e/m cancels first; only then is e/m extracted.
2. **Orbit and orbital are not interchangeable.** An orbit is a path; an orbital is a quantum state/wavefunction.
3. **Exact momentum does not require many wavelengths.** Exact momentum corresponds to one exact de Broglie wavelength. Many wavelengths are required to localize a wave packet.
4. **A node is not a forbidden wall.** It is a region where ψ and therefore probability density are zero.
5. **The drawn orbital boundary is not the end of the wavefunction.** It encloses a chosen high fraction of probability.

6. **Bohr is not literally correct for hydrogen.** His hydrogen energies are excellent, but the definite circular trajectory is not the modern description.
7. **Hydrogen $2s$ and $2p$ can have the same energy despite different shapes.** Degeneracy does not mean identical wavefunctions.
8. **Penetration does not mean an entire orbital is simply closer to the nucleus.** It refers to probability entering inner, weakly shielded regions.
9. **$4s$ fills before $3d$, but transition-metal ions usually lose $4s$ first.** Occupancy changes relative orbital energies.
10. **Isoelectronic does not mean same size.** Larger nuclear charge contracts the same electron count more strongly.
11. **Chromium and copper are exceptions to the simplest Aufbau pattern.** Do not force every element into the mnemonic blindly.

43 The whole chapter as one continuous story

43.1 Thomson

Cathode rays reveal universal negatively charged particles. Electric and magnetic deflections first give their speed and then the universal ratio e/m . Neutral atoms require compensating positive charge, leading to Thomson's diffuse-positive model.

43.2 Rutherford

Gold-foil scattering shows that positive charge and almost all atomic mass are concentrated in a tiny nucleus. But classical orbiting electrons should radiate and collapse.

43.3 Hydrogen spectrum and Bohr

Hydrogen emits only discrete frequencies. Bohr postulates stationary states with quantized energies and transitions obeying

$$\Delta E = h\nu.$$

This explains the Rydberg spectrum of hydrogen and one-electron species.

43.4 de Broglie

Electrons also possess wave character:

$$\lambda = \frac{h}{p}.$$

The purely classical particle picture becomes incomplete.

43.5 Heisenberg

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

A sharply defined classical trajectory is not a physically available atomic state.

43.6 Schrodinger and Born

Replace trajectories with wavefunctions:

$$\hat{H}\psi = E\psi.$$

Physical boundary conditions select allowed states and energies. The spatial probability density is

$$|\psi|^2.$$

43.7 Orbitals and quantum numbers

Allowed one-electron states are organized by

$$n, l, m_l, m_s.$$

These give shells, subshells, shapes, orientations and spin.

43.8 Multi-electron atoms

Electron-electron repulsion introduces shielding, penetration and l -dependent orbital energies.

43.9 Aufbau + Pauli + Hund

These rules organize the ground-state occupation of available orbital states and produce the electronic configurations that underpin periodicity and bonding.

The chapter's deepest transition is from asking "What path does the electron follow?" to asking "What quantum state describes the electron, and what probabilities does that state predict?"

44 Formula sheet

Concept	Formula
Atomic composition	$A = Z + N, N = A - Z$
Electromagnetic wave	$c = \nu\lambda$
Photon energy	$E = h\nu = hc/\lambda$
Rydberg equation	$\tilde{\nu} = R_H(1/n_f^2 - 1/n_i^2)$
Electric deflection, small angle	$\theta \approx eEl/(mv^2)$
Magnetic deflection, small angle	$\phi \approx eBl/(mv)$
Thomson speed from deflections	$v = (E/B)(\phi/\theta)$; if $\phi = \theta$, $v = E/B$
Bohr angular momentum	$mvr = nh/(2\pi) = n\hbar$
Bohr radius, H-like ion	$r_n = 52.9 n^2/Z \text{ pm}$
Hydrogen energy	$E_n = -13.6/n^2 \text{ eV}$
Hydrogen-like energy	$E_n = -2.18 \times 10^{-18} Z^2/n^2 \text{ J}$
de Broglie relation	$\lambda = h/p = h/(mv)$ for nonrelativistic motion
Uncertainty relation	$\Delta x \Delta p_x \geq h/(4\pi) = \hbar/2$
Schrodinger equation	$\hat{H}\psi = E\psi$
Probability in volume element	$dP = \psi ^2 dV$
Orbitals in subshell	$2l + 1$
Orbitals in shell	n^2
Maximum electrons in shell	$2n^2$
Radial nodes	$n - l - 1$
Angular nodes	l

Total nodes

 $n - 1$

Useful constants:

$$h = 6.626 \times 10^{-34} \text{ J s}, \quad c \approx 3.00 \times 10^8 \text{ m s}^{-1},$$

$$e = 1.602 \times 10^{-19} \text{ C}, \quad m_e = 9.11 \times 10^{-31} \text{ kg}.$$

These modern constants are useful for calculations, but remember the historical distinction: Thomson did not know e and m_e separately when he first determined e/m .

45 What to retain from Chapter 4

If the details fade, keep this conceptual spine:

1. Atomic positive charge and almost all mass are concentrated in the nucleus.
2. Z counts protons; $A = Z + N$.
3. Thomson's deflection experiment measured e/m without prior knowledge of e or m : the speed was obtained by combining electric and magnetic deflections.
4. Hydrogen's discrete spectrum reveals discrete energy differences.
5. Bohr successfully quantized hydrogen energies but retained an incorrect literal orbit picture.
6. Electrons have wave-particle character.
7. Quantum uncertainty prevents a perfectly defined classical electron trajectory.
8. Schrodinger's equation plus physical boundary conditions gives allowed wavefunctions and energies.
9. $|\psi|^2$ is a probability density; integrating it over a region gives probability.
10. An orbital is a quantum state, not a path.
11. n, l, m_l, m_s organize electron states; spin is separate from the spatial Schrodinger quantum numbers.
12. s, p, d, f are subshells containing 1, 3, 5 and 7 orbitals.
13. Multi-electron orbital energies depend on electron-electron interaction, shielding and penetration.
14. The $n + l$ rule is a practical guide, not an immutable energy law.
15. Aufbau gives the practical ground-state filling scheme; Pauli limits occupancy; Hund governs filling of degenerate orbitals.
16. Chromium and copper are important exceptions to the simplest Aufbau prediction.
17. Transition-metal cations generally lose $4s$ electrons before $3d$ because the relative orbital energies change with occupancy.
18. Isoelectronic species have the same electron count but become smaller as nuclear charge increases.

Chapter 4 status

Covered in our lessons, conceptual detours and mastery practice:

- subatomic particles and the experiments that established them;

- atomic number, mass number, isotopes, isobars and isotones;
- Rutherford's model and its failure;
- electromagnetic radiation and hydrogen line spectra;
- Bohr's model, its derivation, energy levels, radii and hydrogen spectrum;
- limitations of Bohr's theory;
- de Broglie matter waves and Heisenberg uncertainty;
- Schrodinger's quantum-mechanical model and why quantization emerges;
- orbitals, probability density, quantum numbers, shapes and nodes;
- orbital energies, shielding, penetration and the $n + l$ rule;
- Aufbau, Pauli and Hund;
- electronic configurations, Cr/Cu exceptions, transition-metal ions and isoelectronic species;
- integrated mastery problems connecting the whole chapter.

Chapter 4: conceptually complete.